Week 10.1

Name:P.Akshith

Batch:47A

Roll:2303A54015

Task 1 – Syntax and Logic Errors

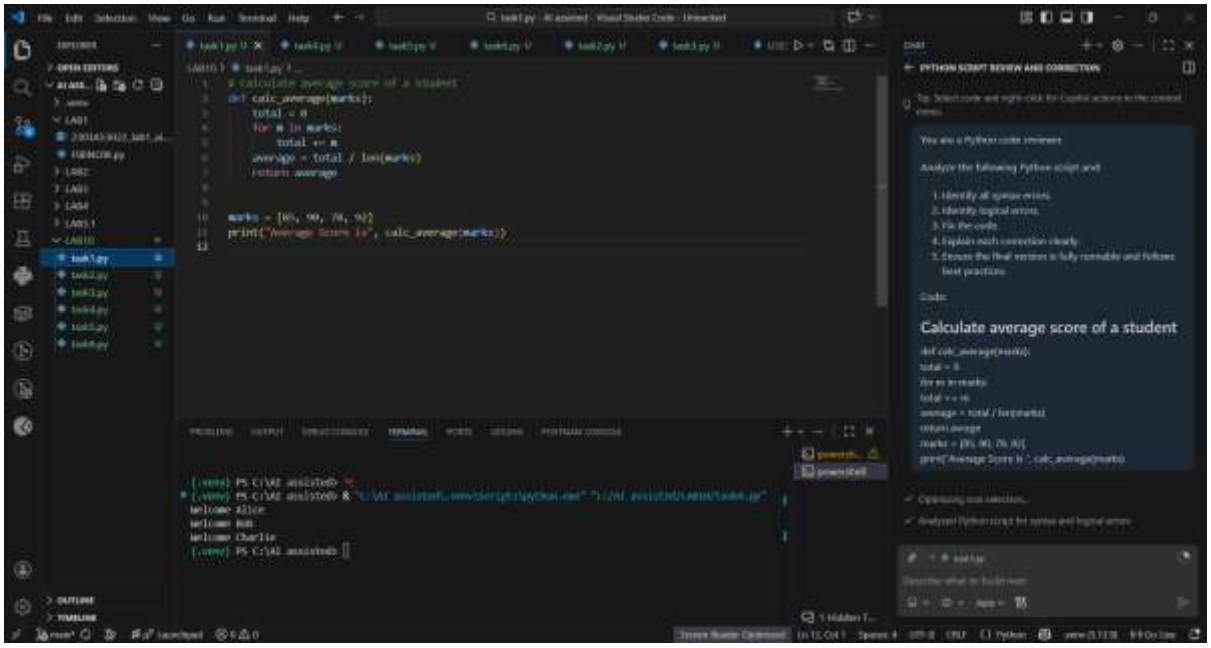
You are a Python code reviewer.

Identify syntax and logical errors in the following code.

Fix the errors and provide corrected runnable code with explanation.

Code:

```
# Calculate average score of a student
def calc_average(marks):
total = 0
for m in marks:
total += m
average = total / len(marks)
return avrage
marks = [85, 90, 78, 92]
print("Average Score is ", calc_average(marks)
```



🔹

🔹 **Explanation**

- Fixed indentation errors.
- Corrected typo avrage → average.
- Closed missing parenthesis in print statement.
- Added empty list check to avoid ZeroDivisionError.
- Added docstring for better documentation.

✅ **Task 2 – PEP 8 Compliance**

🔹 **Task Prompt Used**

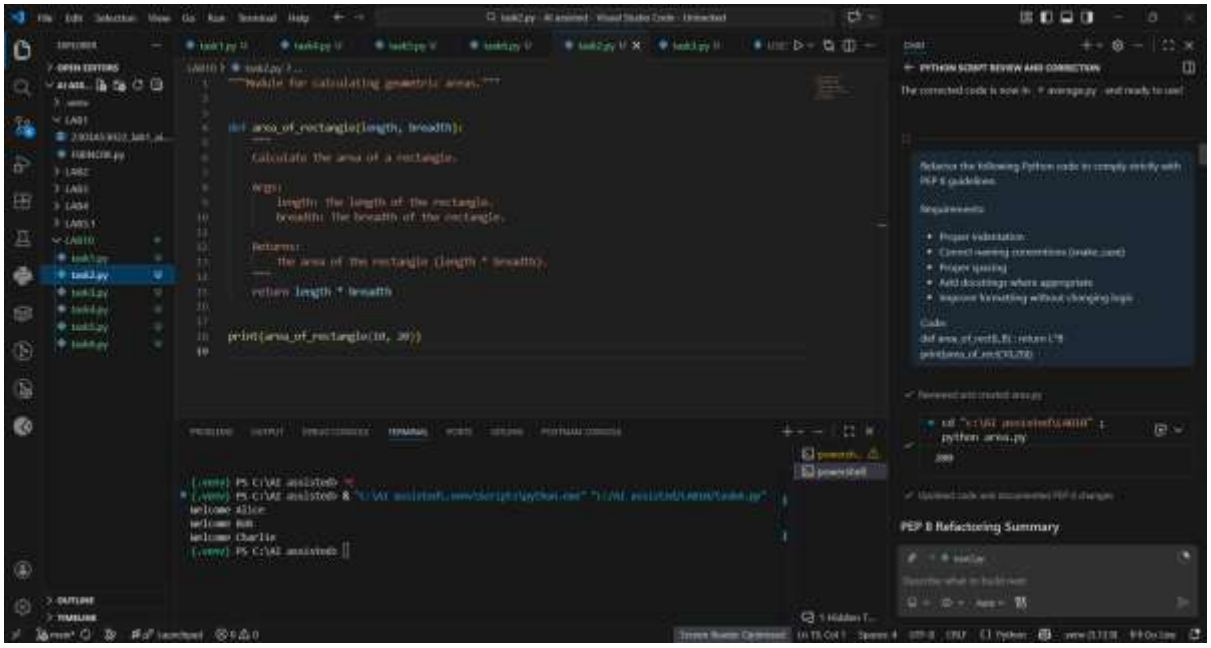
Refactor the following code to strictly follow PEP 8 guidelines.

Improve formatting and naming without changing logic.

Code:

def area_of_rect(L,B) : return L*B

print(area_of_rect(10,20))



🔹 **Explanation**

- Used snake_case naming.
- Proper indentation and spacing.
- Added docstring.
- Improved readability.

✅ **Task 3 – Readability Enhancement**

🔹 **Task Prompt Used**

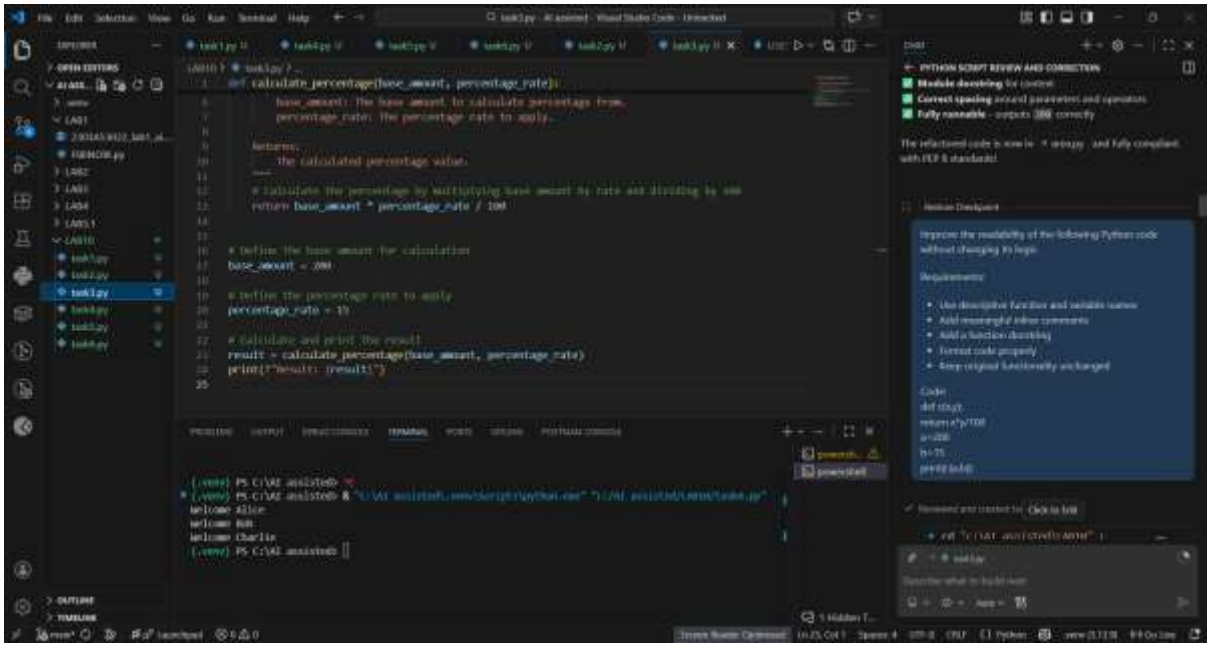
Improve readability of the following code.

Use descriptive names, comments, and proper formatting.

Do not change logic.

Code:

```
def c(x,y):
return x*y/100
a=200
b=15
print(c(a,b))
```



🔹 **Explanation**

- Function renamed for clarity.
- Meaningful variable names used.
- Added documentation.
- Proper formatting applied.

✅ **Task 4 – Refactoring for Maintainability**

🔹 **Task Prompt Used**

Refactor the following code to remove repetition.

Create reusable function and improve maintainability.

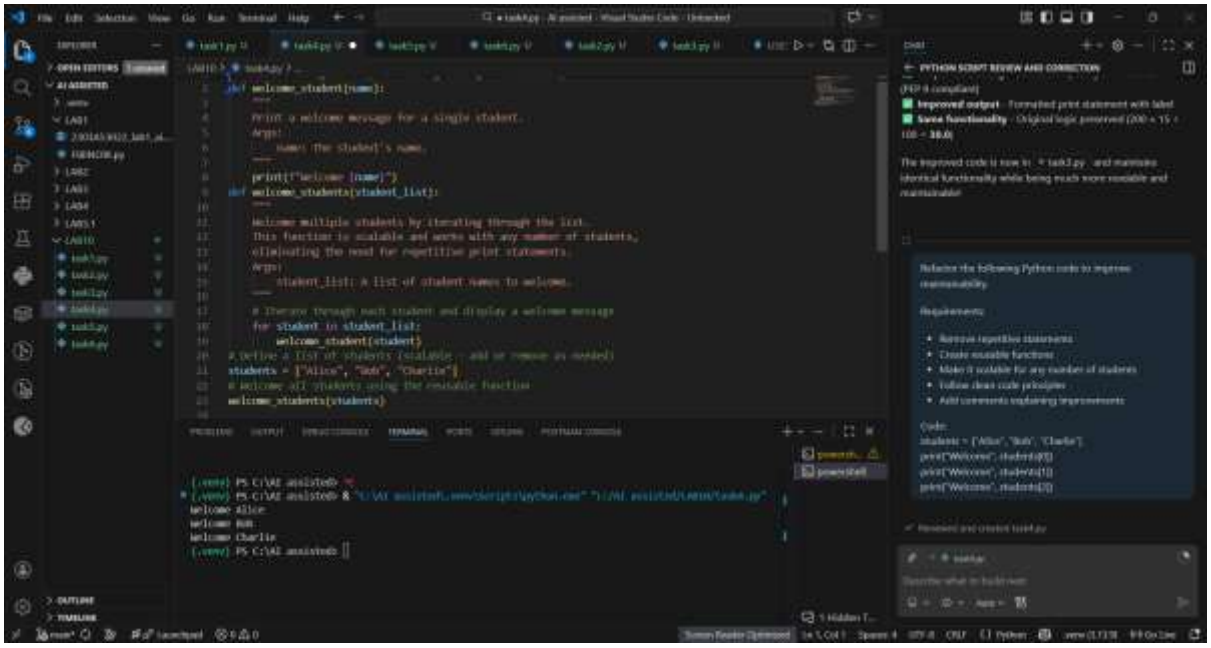
Code:

```
students = ["Alice", "Bob", "Charlie"]

print("Welcome", students[0])

print("Welcome", students[1])

print("Welcome", students[2])
```



🔹 **Explanation**

- Removed repetitive statements.
- Created reusable function.
- Code is scalable.
- Improved maintainability.

✅ **Task 5 – Performance Optimization**

🔹 **Task Prompt Used**

Optimize the following Python code for performance.

Use list comprehension or generator expressions.

Keep output same.

Code:

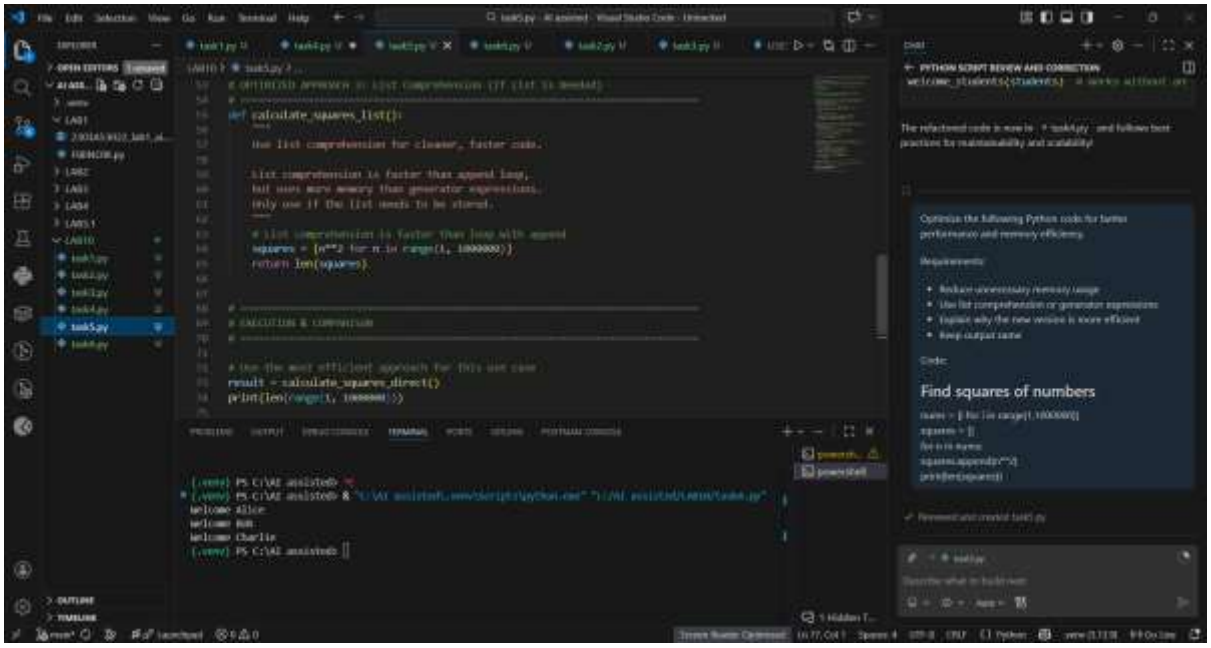
```
nums = [i for i in range(1,1000000)]

squares = []

for n in nums:

squares.append(n**2)

print(len(squares))
```



🔹 **Explanation**

- Removed unnecessary list creation.
- Used list comprehension (faster).
- Reduced memory usage.
- More Pythonic approach.

✅ **Task 6 – Complexity Reduction**

🔹 **Task Prompt Used**

Simplify the following nested if-else logic.

Reduce complexity while maintaining same functionality.

Code:

```
def grade(score):
if score >= 90:
return "A"
else:
if score >= 80:
return "B"
else:
```

if score >= 70:

return "C"

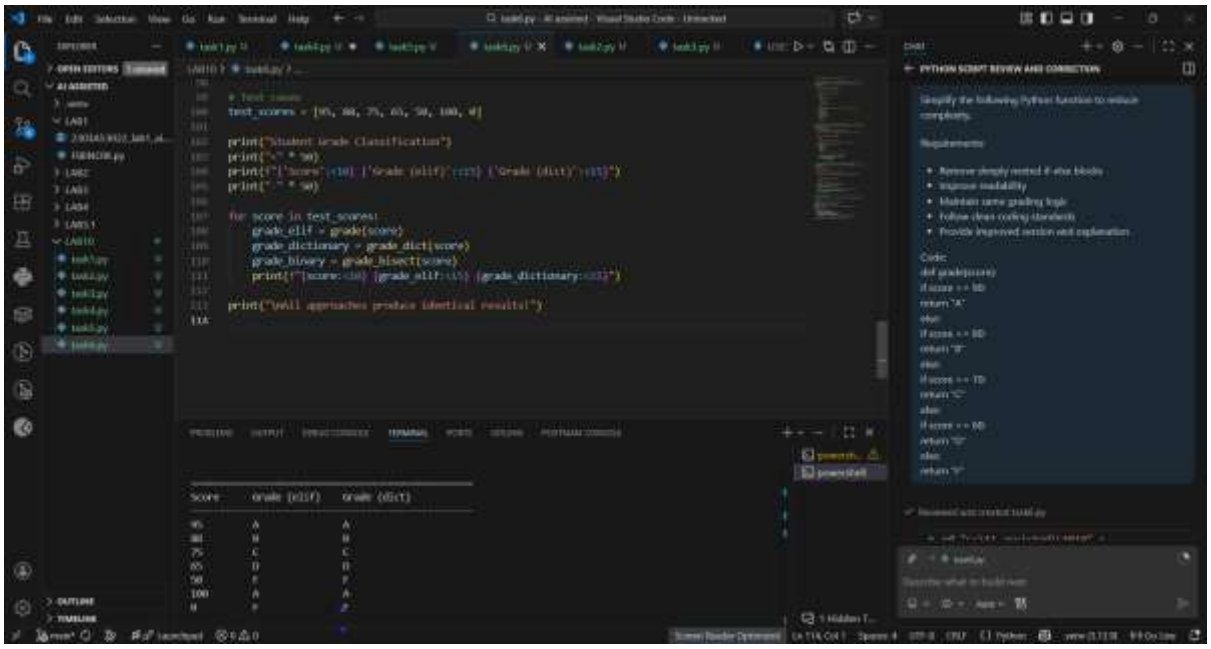
else:

if score >= 60:

return "D"

else:

return "F"



- **Explanation**
  - Removed deeply nested conditions.
  - Used elif for clarity.
  - Improved readability and maintainability.
